

Experiment No. 1

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File Path: file:///Users/ojaswinithote/Desktop/Javascript/EXP%201/index1.html

Experiment Title

Demonstration of Inline, Internal and External JavaScript, Console Methods and Student Information Webpage

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

index.html — Department Home Page

```
<!DOCTYPE html>
<html>
<head>
  <title>Welcome to SIT!</title>

  <style>
    body{
      font-family: Arial, sans-serif;
      background-color: lightpink;
    }

    .box{
      width: 450px;
      margin: 30px auto;
      padding: 20px;
      border: 2px solid black;
      border-radius: 10px;
      background-color: white;
    }

    h1, h2{
      text-align: center;
    }

    input[type="text"],
    input[type="number"],
    select{
      width: 100%;
      padding: 5px;
      margin-top: 5px;
```

```

        margin-bottom: 10px;
        box-sizing: border-box;
    }

    button{
        padding: 8px 15px;
        background-color: blue;
        color: white;
        border: none;
        cursor: pointer;
    }

    button:hover{
        background-color: darkblue;
    }

    #output{
        margin-top: 20px;
        padding: 10px;
        border: 1px solid black;
        background-color: #f8f8f8;
    }
</style>

<!-- Internal JavaScript -->
<script>
    function showWelcome() {

        let name = document.getElementById("name").value;
        let age = document.getElementById("age").value;
        let cgpa = document.getElementById("cgpa").value;
        let email = document.getElementById("email").value;
        let phone = document.getElementById("phone").value;

        // Get selected semester
        let semester = document.querySelector('input[name="semester"]:checked').value;

        // Get selected section
        let section = document.getElementById("section").value;

        document.getElementById("output").innerHTML =
            "<h2>Student Information</h2>" +
            "<p><b>Name:</b> " + name + "</p>" +
            "<p><b>Age:</b> " + age + "</p>" +
            "<p><b>CGPA:</b> " + cgpa + "</p>" +
            "<p><b>Semester:</b> " + semester + "</p>" +
            "<p><b>Section:</b> " + section + "</p>" +
            "<p><b>Email:</b> " + email + "</p>" +
            "<p><b>Phone Number:</b> " + phone + "</p>";

        console.log("Name: " + name);
        console.info("Age: " + age);
        console.log("CGPA: " + cgpa);
        console.log("Semester: " + semester);
        console.log("Section: " + section);
        console.log("Email: " + email);
        console.log("Phone: " + phone);

        console.warn("This is a warning message");
        console.error("This is an error message");
    }
</script>

```

```

    }
  </script>
</head>

<body>

<div class="box">

  <h1>Welcome to SIT!</h1>
  <h2>Student Information</h2>

  <label>Enter Name:</label><br>
  <input type="text" id="name" placeholder="Enter Name"><br>

  <label>Enter Age:</label><br>
  <input type="number" id="age" placeholder="Enter Age"><br>

  <label>Enter CGPA:</label><br>
  <input type="text" id="cgpa" placeholder="Enter CGPA"><br>

  <label>Select Semester:</label><br>

  <input type="radio" name="semester" value="Semester 1" checked> Semester 1<br>
  <input type="radio" name="semester" value="Semester 2"> Semester 2<br>
  <input type="radio" name="semester" value="Semester 3"> Semester 3<br>
  <input type="radio" name="semester" value="Semester 4"> Semester 4<br>
  <input type="radio" name="semester" value="Semester 5"> Semester 5<br>
  <input type="radio" name="semester" value="Semester 6"> Semester 6<br>
  <input type="radio" name="semester" value="Semester 7"> Semester 7<br>
  <input type="radio" name="semester" value="Semester 8"> Semester 8<br><br>

  <label>Select Section:</label><br>

  <select id="section">
    <option value="Section A">Section A</option>
    <option value="Section B">Section B</option>
    <option value="Section C">Section C</option>
    <option value="Section D">Section D</option>
  </select>

  <label>Enter Email ID:</label><br>
  <input type="text" id="email" placeholder="Enter Email"><br>

  <label>Enter Phone Number:</label><br>
  <input type="number" id="phone" placeholder="Enter Phone Number"><br><br>

  <button onclick="showWelcome(); greet();">
    Submit
  </button>

  <hr>

  <div id="output"></div>

</div>

<script src="script.js"></script>

</body>
</html>

```

Description:

- Displays the department homepage.
- Contains navigation to the Student Information page.
- Uses an external JavaScript file (script.js) for alert messages and console methods.

script.js — External JavaScript

```
function greet() {  
  
    alert("Welcome to SIT, Nagpur!");  
  
    console.table([  
        {  
            Name: "Sample User",  
            Course: "JavaScript"  
        }  
    ]);  
  
    console.time("Execution");  
    console.timeEnd("Execution");  
  
}
```

Description:

- Defines the greet() function.
- Displays a welcome alert.
- Demonstrates JavaScript console methods: console.table(), console.time(), and console.timeEnd().

website.html — Student Information Page (Internal + Inline JavaScript)**Description:**

- Accepts student name and age.
- Validates user input.
- Generates a dynamic student details table.
- Uses Internal JavaScript for form validation and table generation.
- Uses Inline JavaScript through the button's onclick event.
- Uses External JavaScript by calling greet() from script.js.

Output

- The department homepage loads successfully.
- Clicking View Student Info Page opens the student information form.
- A welcome alert is displayed.
- Student information is shown in a formatted table.
- Browser console displays console.log(), console.info(), console.warn(), console.error(), console.table(), console.time(), and console.timeEnd().

Screenshot

Vehicle Registration Validator x Welcome to SIT! x + Ask Gemini

File /Users/ojaswinithote/Desktop/Javascript/EXP%201/index1.html

Student Information

Enter Name:
Ojaswini Thote

Enter Age:
20

Enter CGPA:
9

Select Semester:
☐ Semester 1
☐ Semester 2
☐ Semester 3
☒ Semester 4
☐ Semester 5
☐ Semester 6
☐ Semester 7
☐ Semester 8

Select Section:
Section A

Enter Email ID:
ojaspt@gmail.com

Enter Phone Number:
8767064228

Submit

Student Information

Name: Ojaswini Thote
Age: 20
CGPA: 9
Semester: Semester 4
Section: Section A
Email: ojaspt@gmail.com
Phone Number: 8767064228

Screenshot 2:

Welcome to SIT! x + Ask Gemini

File /Users/ojaswinithote/Desktop/index.html

Welcome to SIT!

Student Information

Enter Name:
Enter Name

Enter Age:
Enter Age

Enter CGPA:
Enter CGPA

Select Semester:
☒ Semester 1
☐ Semester 2
☐ Semester 3
☐ Semester 4
☐ Semester 5
☐ Semester 6
☐ Semester 7
☐ Semester 8

Select Section:
Section A

Enter Email ID:
Enter Email

Enter Phone Number:
Enter Phone Number

Submit

Case Study Title

Student Information Webpage Connected to the Department Homepage using JavaScript

Case Study Screenshot:

<https://github.com/user-attachments/assets/27d1ab43-29ef-4da1-948c-a75cd0ad70d2>

Case Study Description

This practical demonstrates the implementation of all three methods of including JavaScript within a single web application.

JavaScript Type	Where Used	Purpose
Inline JavaScript	onclick="showWelcome(); greet();"	Calls JavaScript functions when the button is clicked
Internal JavaScript	<script> block inside website.html	Validates input and dynamically generates the student information table
External JavaScript	script.js	Displays a welcome alert and demonstrates console methods

Output Summary

- Department homepage opens successfully.
- Navigation to the student information page works correctly.
- Student details are displayed dynamically after form submission.
- Alert message appears using external JavaScript.
- Console methods execute successfully.
- Input validation prevents empty submissions.

Screenshot

The screenshot shows a web browser window with the title 'Welcome to SIT!'. The address bar displays the file path: `/Users/ojaswinithote/Desktop/Javascript/EXP%201/index1.html`. The browser's developer tools are open, showing the 'File' tab. The main content area features a form titled 'Welcome to SIT!' and 'Student Information'. The form includes the following fields and controls:

- Enter Name:** A text input field containing 'Ojaswini Thote'.
- Enter Age:** A text input field containing '20'.
- Enter CGPA:** A text input field containing '9'.
- Select Semester:** A radio button group with options for Semesters 1 through 8. 'Semester 4' is selected.
- Select Section:** A dropdown menu showing 'Section A'.
- Enter Email ID:** A text input field containing 'ojaspt@gmail.com'.
- Enter Phone Number:** A text input field containing '8767064228'.
- Submit:** A blue button labeled 'Submit'.

Result / Conclusion

The experiment was completed successfully. The implementation demonstrated the use of Inline, Internal, and External JavaScript within a single web application. Various console methods including `console.log()`, `console.info()`, `console.warn()`, `console.error()`, `console.table()`, `console.time()`, and `console.timeEnd()` were successfully executed. The project effectively showcases JavaScript event handling, form validation, DOM manipulation, and dynamic content generation through a Student Information Webpage connected to the Department Homepage.